

PROBLEM STATEMENT

Experiment 2 — Reinforcement Learning: Q-Learning

4×4 Grid World Environment

Problem Statement

Implement Q-Learning for a simple 4×4 Grid World environment and observe the agent's learned policy.

(a) Environment Definition

Define the environment with states and four actions: Up, Down, Left and Right. Use a reward structure of +10 for reaching the Goal, −1 for each normal step, and −5 for entering an obstacle. Initialize the Q-table with zeros and state the hyperparameters α , γ and ϵ .

(b) Q-Learning Training

Run the Q-Learning algorithm for at least 100 episodes. Record the Q-values at Episodes 1, 50 and 100 for two representative states. Show how the Q-table evolves during learning.

(c) Learned Policy and Convergence

Extract and display the final learned policy by selecting the optimal action at every state. Plot cumulative reward per episode and justify the convergence behaviour.

Expected Outcome

The agent should learn a policy that reaches the goal while avoiding unnecessary steps and obstacle penalties. The Q-table should gradually improve as experience is accumulated.